



Introduction

This study explored the potential of Large Language Models (LLMs) like Mistral and Llama-2 to enhance Network Intrusion Detection Systems (NIDS) capabilities. By fine-tuning these models on network traffic data, we aimed to improve intrusion detection performance and address the limitations of traditional NIDS, which face challenges in keeping up with evolving cyber threats [1].

Method

- Mistral and Llama-2 were the LLMs used in this study.
- The CIDDS dataset was preprocessed using the NTFA tool and python scripts for fine-tuning.
- Fine-tuning was done using libraries like transformers, PEFT and tokenizer.
- Standard metrics such as accuracy, precision, recall, and F1 score were used.



Evaluation Metrics	LLama-2	Mistral
Accuracy	0.53	0.89
Precision	0.96	0.97
Recall	0.54	0.86
F1 Score	0.69	0.91

Table 1. Fine-tuned Llama-2 model vs. Mistral model

Results and Conclusion

- Based on the findings of this study, Mistral shows promise as a candidate for LLM-NIDS, outperforming Llama-2 in several key areas.
- Fine-tuning on a balanced dataset was crucial for improving model performance.
- Mistral, when fine-tuned on a dataset with only 2 types of intrusions, demonstrated strong generalization to unseen data.
- Further research with larger and more diverse datasets could provide deeper insights into the effectiveness and scalability of LLM-NIDS

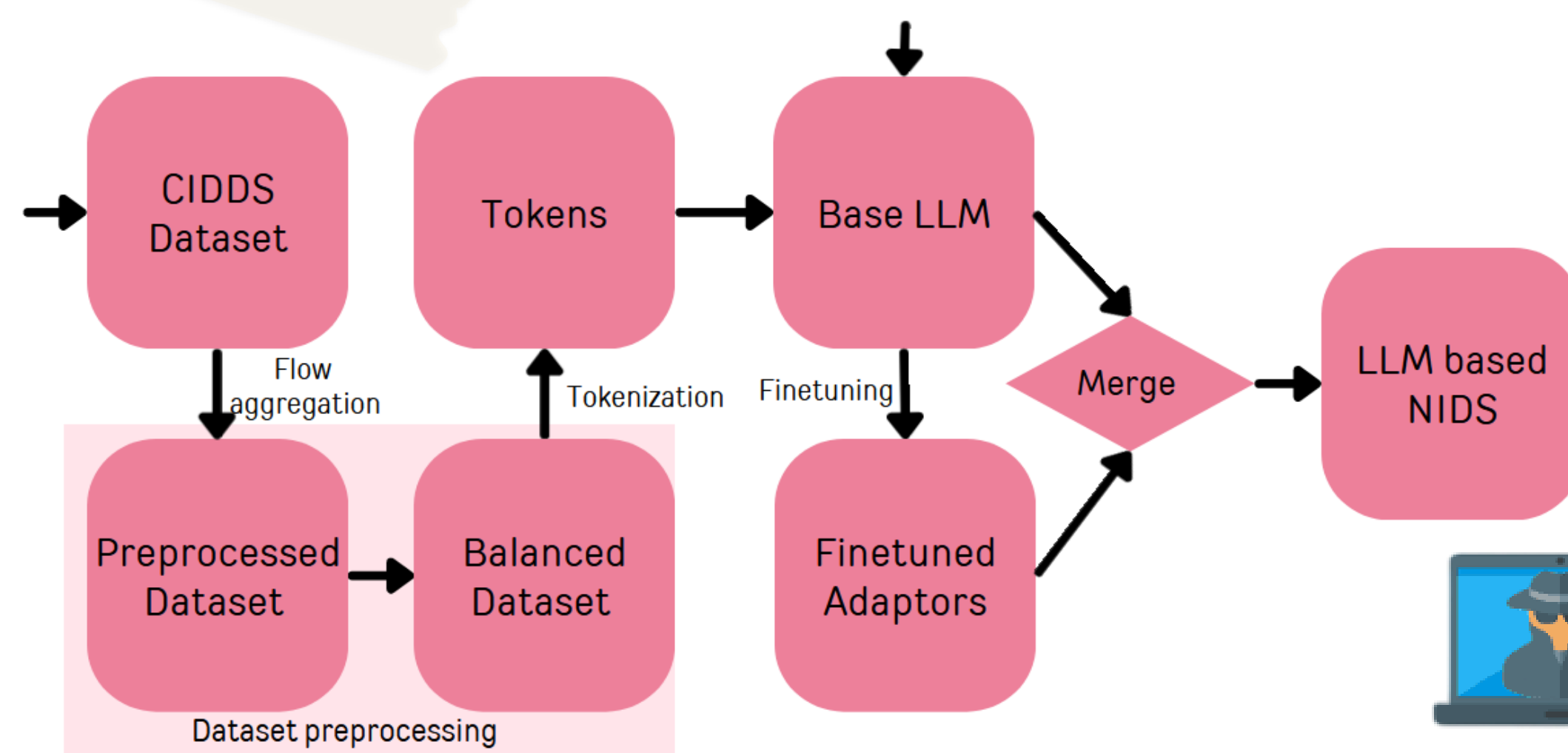
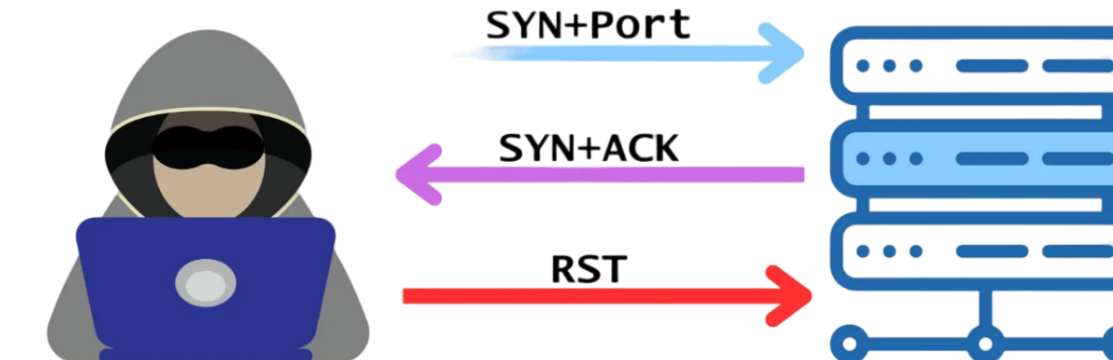


Figure 1. Workflow diagram

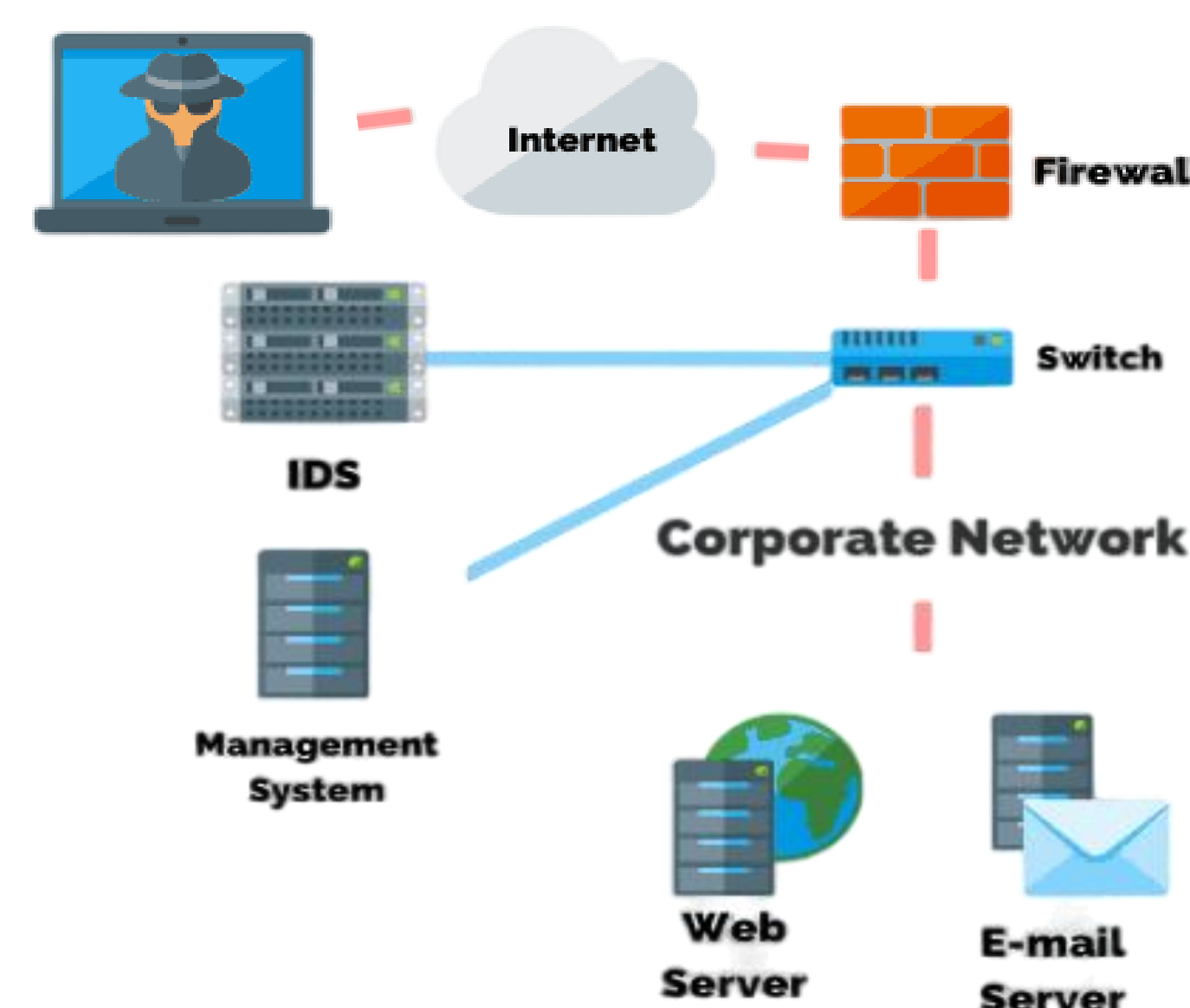


Figure 2. An IDS in a network infrastructure [2]

References

- [1] Abdallah, E. Et Al. (2022). Intrusion detection systems using supervised machine learning techniques: A survey.
- [2] Swanagan, M. (2023). Intrusion detection vs prevention systems: What's the difference? purplesec.us, accessed on 28th Nov, 2023.



Acknowledgments

I thank Professor Kayvan for his support and guidance throughout this project.